

Improving Appointment Attendance Through Data-Driven Decision Support

HealthConnect Clinic Experience Lab

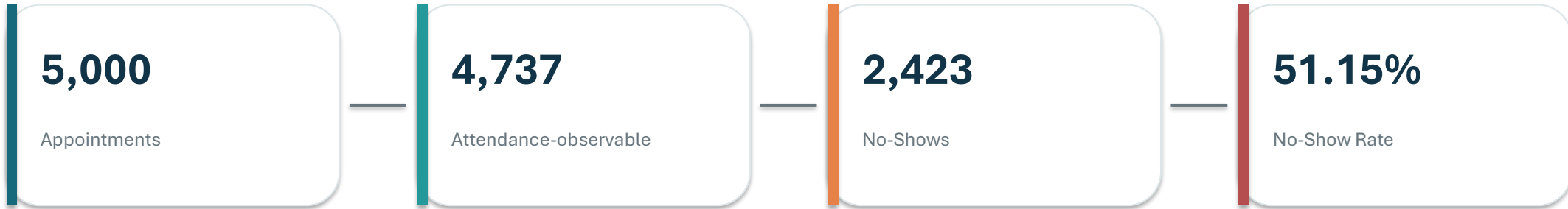
Data Analytics Track

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Missed appointments are a major operational challenge



DECISION QUESTION

Where should HealthConnect start if it wants to test attendance support and how should success be measured?

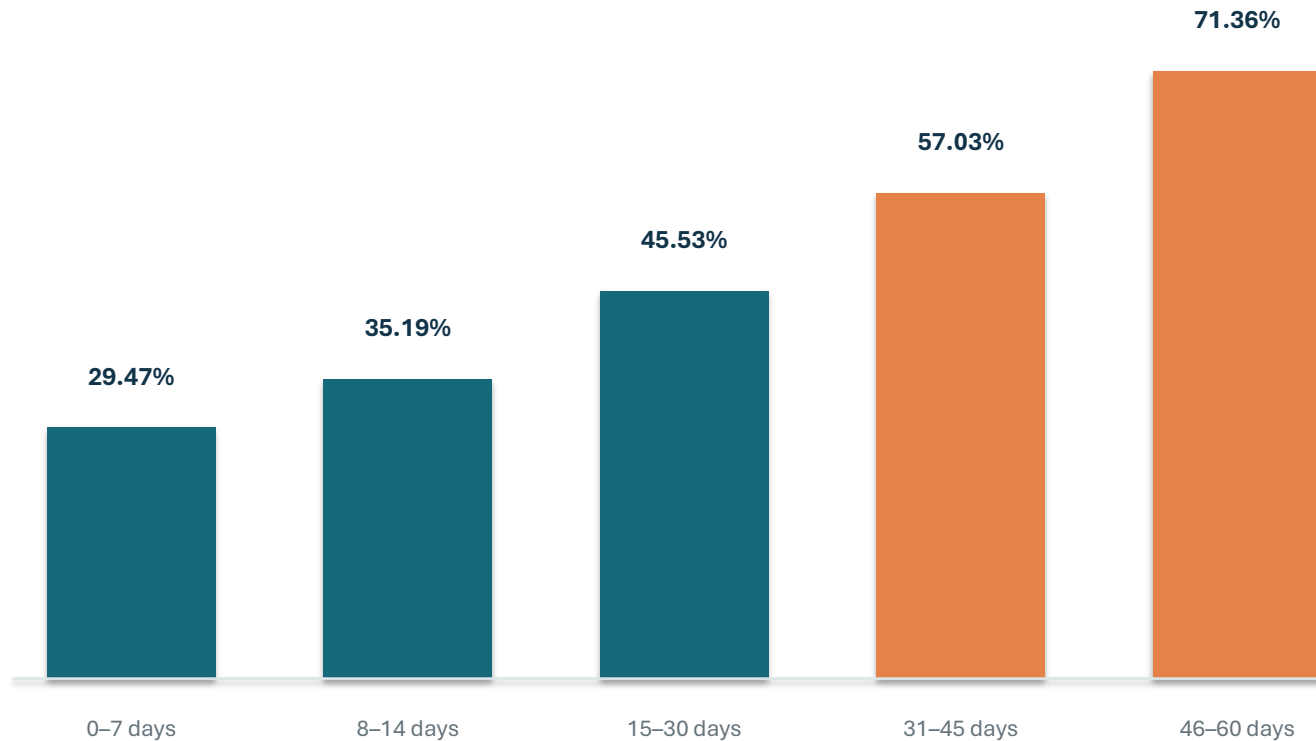
No-Show Rate uses the attendance-observable population (Attended + No-Show).

From appointment data to an actionable decision



Python • SQL • Power BI

Longer booking lead times show the strongest no-show differentiation



Booking Lead Time (days before appointment)

39.13% → 63.95%

0-30 days vs 31-60 days

+24.82pp

PRIORITISE

**Start attendance-support
evaluation with longer-lead appointments.**

Recorded history helps refine where to evaluate first

0–30 days · History = 0

n = 1,428



34.52%

0–30 days · History ≥ 1

n = 1,015



45.62%

31–60 days · History = 0

n = 1,317



59.07%

31–60 days · History ≥ 1

n = 977



70.52%

Lead Time
sets the priority.

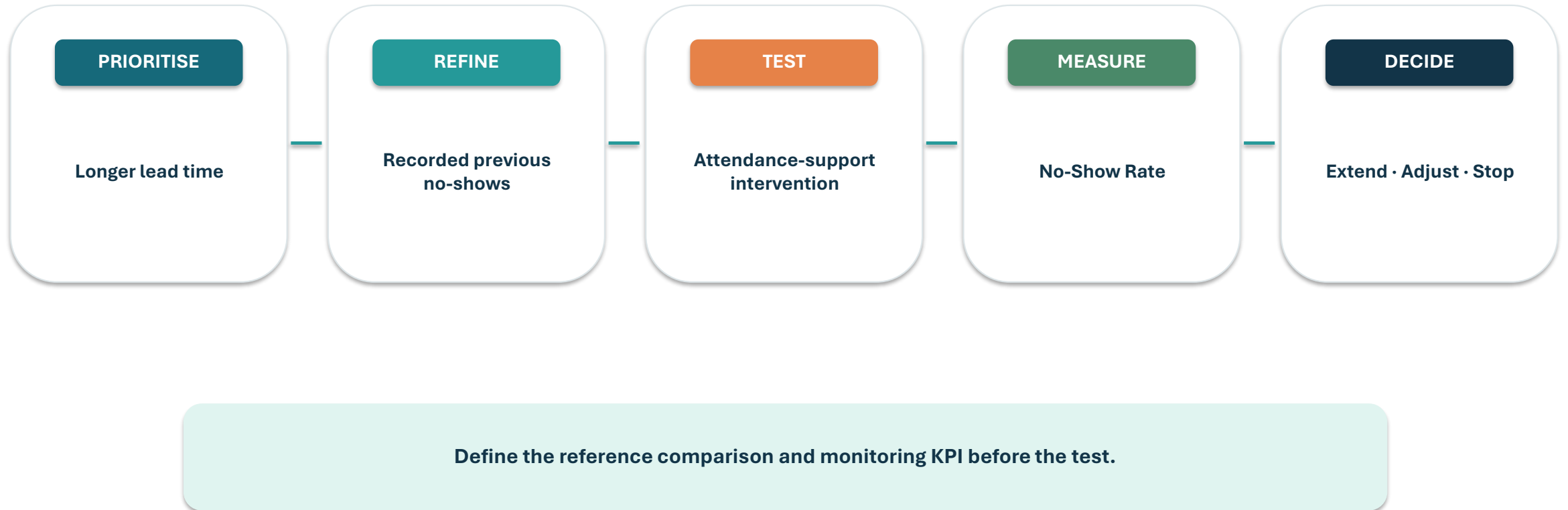
Recorded history
refines it.

REFINE

Use recorded previous no-shows as complementary group

Observed groups — not individual risk predictions.

The evidence supports a test-and-measure strategy — not immediate rollout



Cross-track testing changed two modelling decisions

long_lead_followup

REMOVED

No meaningful incremental predictive value demonstrated.

distance_to_clinic_km

RETAINED IN MODEL

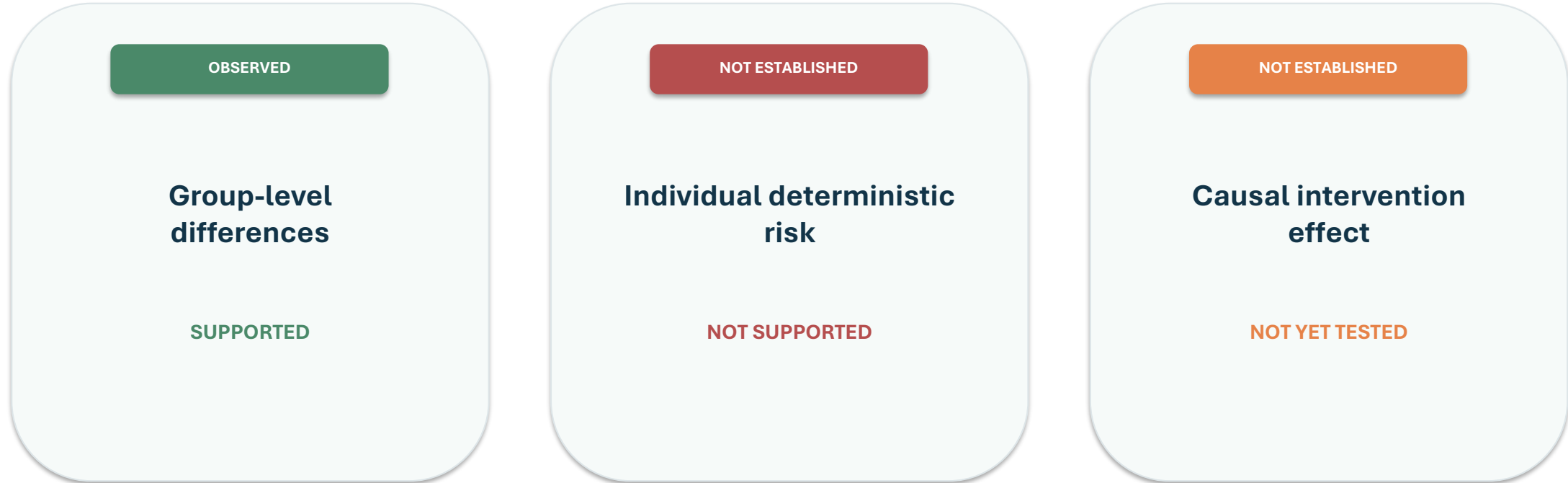
Modest but consistent multivariable contribution.

Analytics evidence ≠ Predictive contribution ≠ Causality

PROJECT MANAGEMENT

Handoff provided and integration requested. • No PM feedback received before finalisation.

What the evidence supports — and what it does not



Patient-history consistency is limited • Some Distance and Waiting Time values are missing • Extreme segments are sparse

Start where the evidence is strongest — then measure before scaling



The value of the analysis is not only identifying where no-shows are higher .
It is helping HealthConnect decide where to act first and how to learn from that action.

THANK YOU